

## INTRODUCTION

- Animal waste from farms is contained in manure lagoons, which can leak and pollute the environment.
- The animal waste contains a large amount of nitrogen and phosphorus.
- Green algae is commonly used in manure lagoons to consume excess nitrogen and phosphorus.
- The excess growth of algae can have negative environmental impacts such as algal blooms which can make bodies of water hypoxic (Nur et. al)
- Repurposing algae can prevent algal blooms and have beneficial impacts on the environment.
- Benefits to the environment may include fewer greenhouse gas emissions and decreased heavy metal pollution. Possible economic impacts would include greater crop yields or lower gas prices (Ma et al.)

### Problem Statement

CAFOs (concentrated animal feeding operations) use lagoons to store animal waste and wastewater. Because of the waste, these lagoons are usually filled with nutrients, heavy metals, and pathogenic microorganisms. It is possible for these lagoons to leak water, which can harm the ecosystem with the flood of extra nutrients-mainly nitrogen and phosphorus. Excess nitrogen and phosphorus can cause major environmental damage. We will design a method to remove these excess nutrients from the ecosystem and the lagoons and put it to good use.

## IDEATION

- Stakeholders:** Stakeholders include farmers with CAFOs who produce algae, farmers who use the algae as fertilizer, consumers who buy the crops, and investors such as farmers or environmentalists.
- Biofertilizer:** Algae is rich with nutrients, most importantly nitrogen and phosphorus, that are essential to plant growth. These attributes improve soil structure, moisture retention, and increase soil Biodiversity along with enzyme activity (Ammar et al.)
- Bioplastics:** Algae’s renewability, biodegradability, and sustainability are useful attributes for bioplastic creation as it is helpful for the environment and does not have to compete with active crops.
- Biofuels:** Algae yields more oil per acre than other crops. This oil can be extracted and synthesized into various fuels.

| Constraints                  | Criteria                   |
|------------------------------|----------------------------|
| Must decrease pollution      | Lower cost = better        |
| Must prevent algal blooms    | Higher profit = better     |
| Must safely absorb nutrients | Less energy usage = better |
|                              | Fast algae growth          |

## EVALUATION OF ALTERNATIVES

- Biofertilizer:
  - Pros: Algae based biofertilizers improved fruit yield by 15.7%-29.6% (Ma et al.) Biosorption and biotransformation are detoxifying mechanisms which can remove heavy metals from soil and water (Ammar et al.)
  - Cons: High energy inputs and high cost to harvest algae.
- Bioplastics:
  - Pros: Reduces microplastic pollution.
  - Cons: Not all bioplastics are biodegradable. Using crops could pose future food security problems (Cheah e al.)
- Biofuels:
  - Pros: Reduces greenhouse gas emissions, they are a renewable energy source, reduces reliance on fossil fuels
  - Cons: The most expensive alternative, vehicle engines optimization, still facing greenhouse gas emissions

Why biofertilizer?  
We found biofertilizers to be more beneficial to society than other options. Increased crop yield can feed more people, lowering food cost.

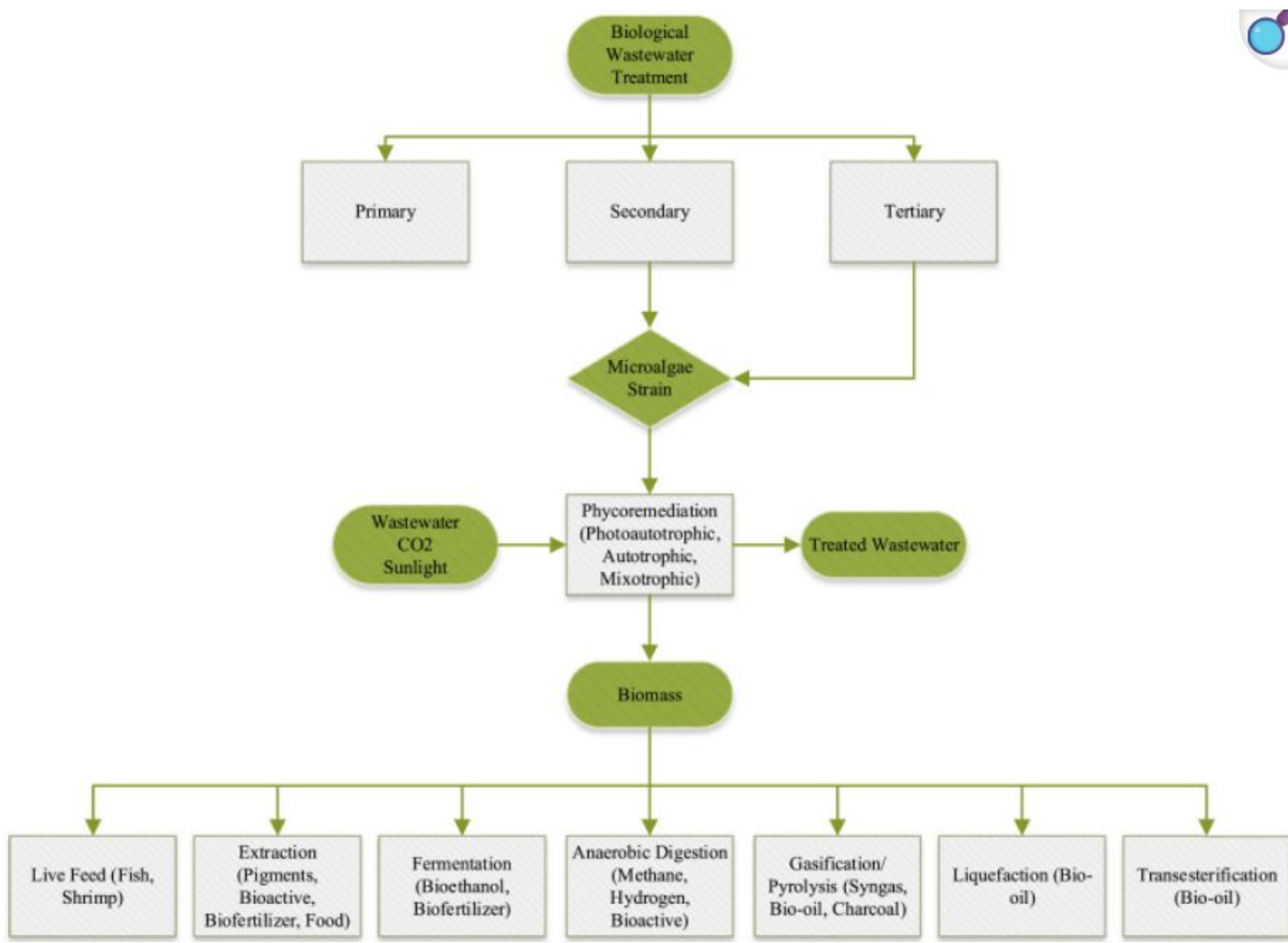


Figure 1. Pathway of algae for other uses

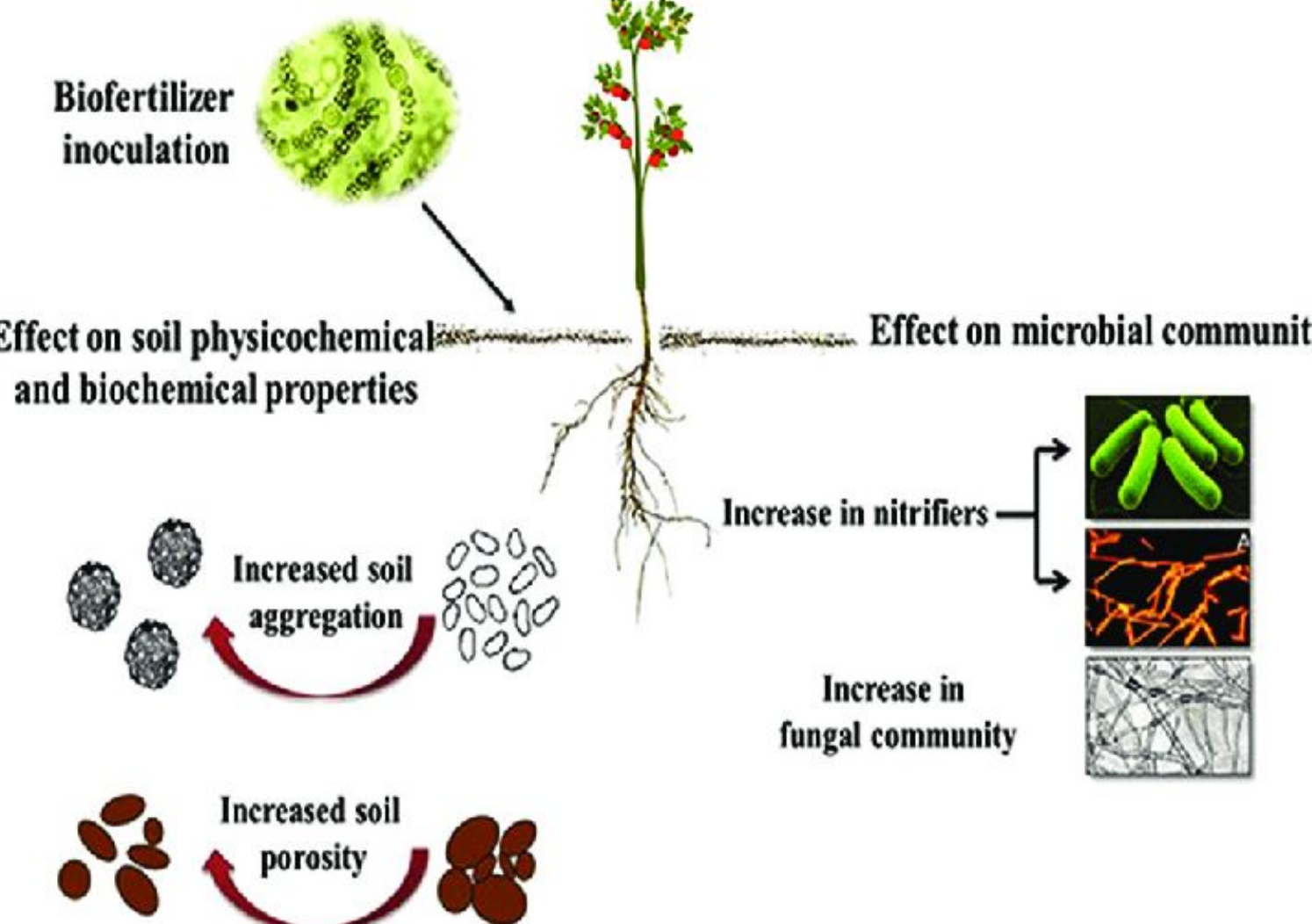


Figure 2: Role of biofertilizers on properties of soil

## DATA COLLECTION & ANALYSIS

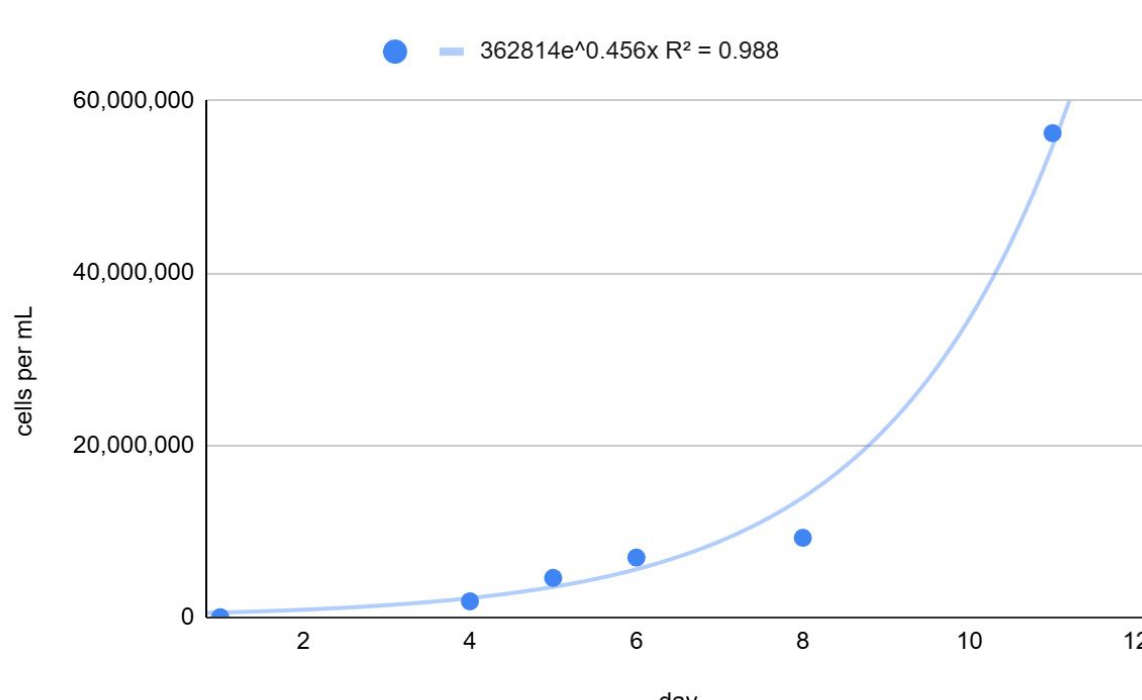


Figure 3: Algae (Chlorella) growth in cells per mL over time.

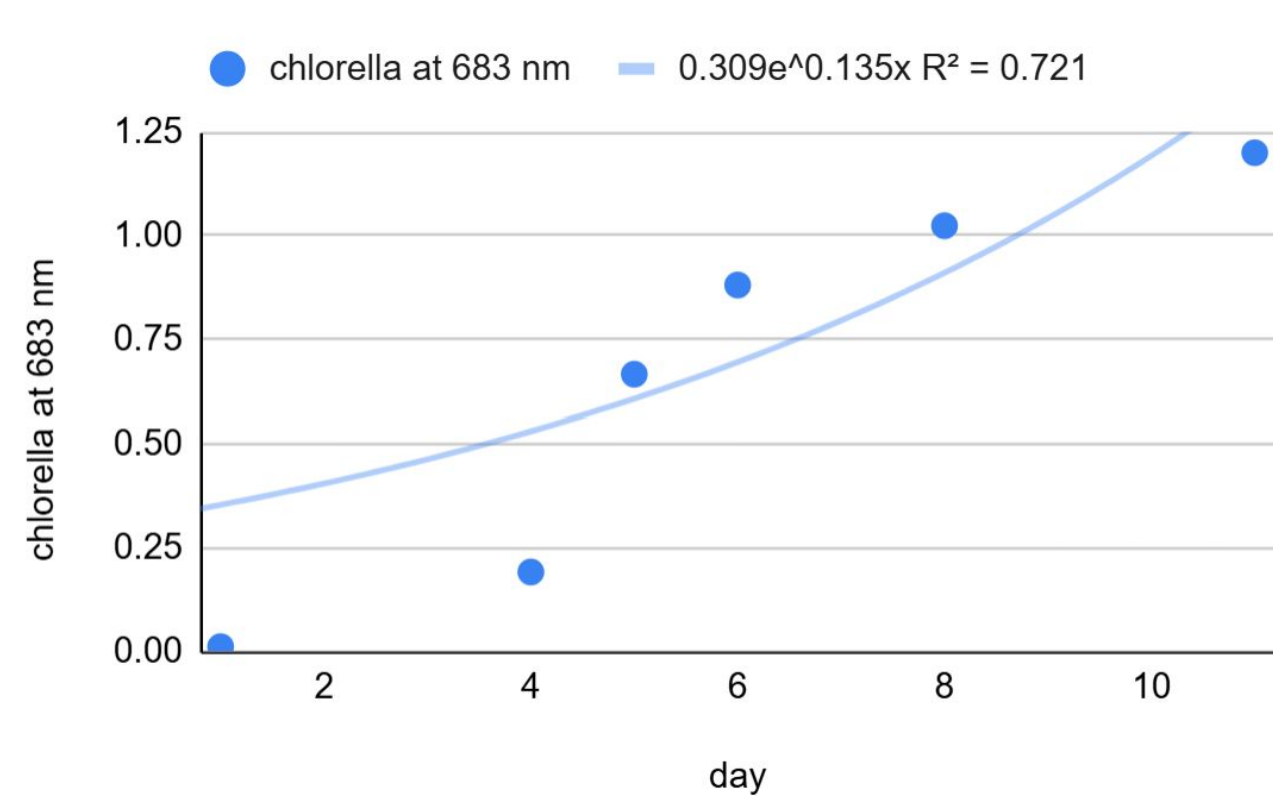


Figure 5: Absorbance of chlorella over time at 683 nm

Analysis: The concentrations of nitrates and phosphates are inversely related to the algal growth which shows the algae absorbed the nitrates and phosphates from the wastewater. As concentration goes down, conductivity goes down because both nitrate and phosphate are ions within the algae solution. Absorbance goes up due to the growth of new cells. The algae removes the ions as it grows.

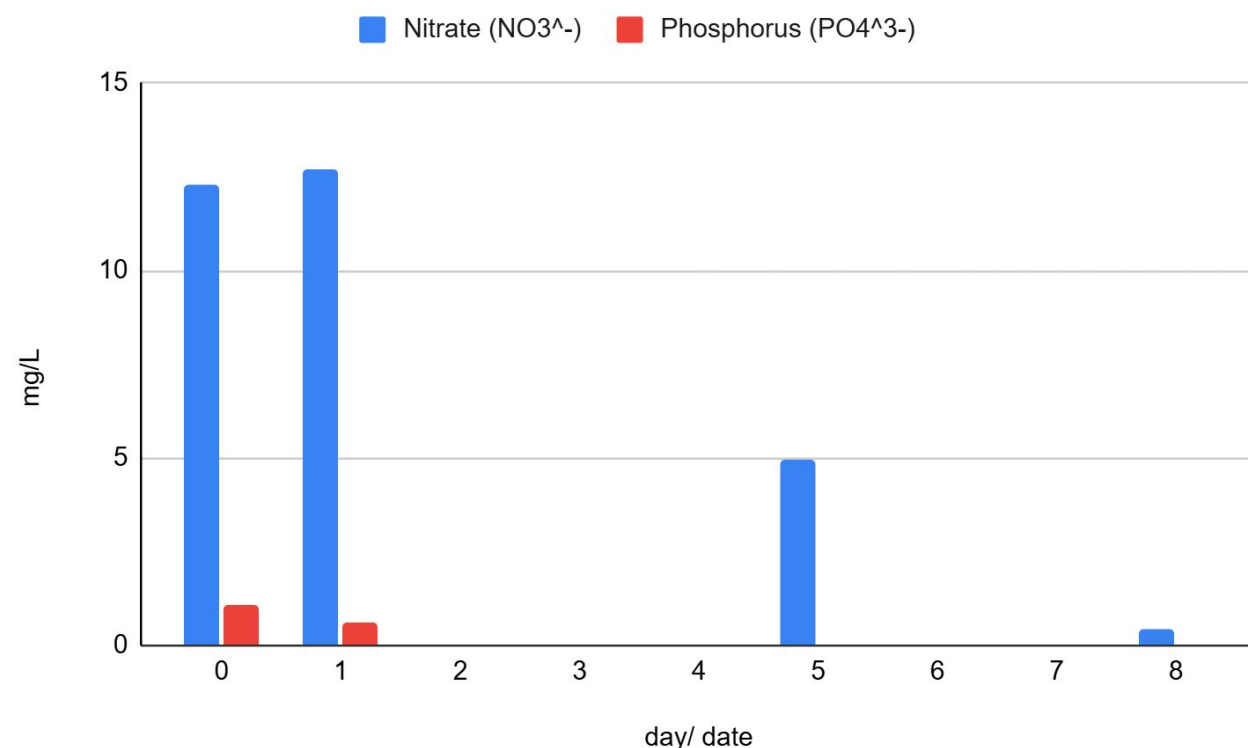


Figure 4: Concentrations of phosphate and nitrate in algae over time. (mg/L)

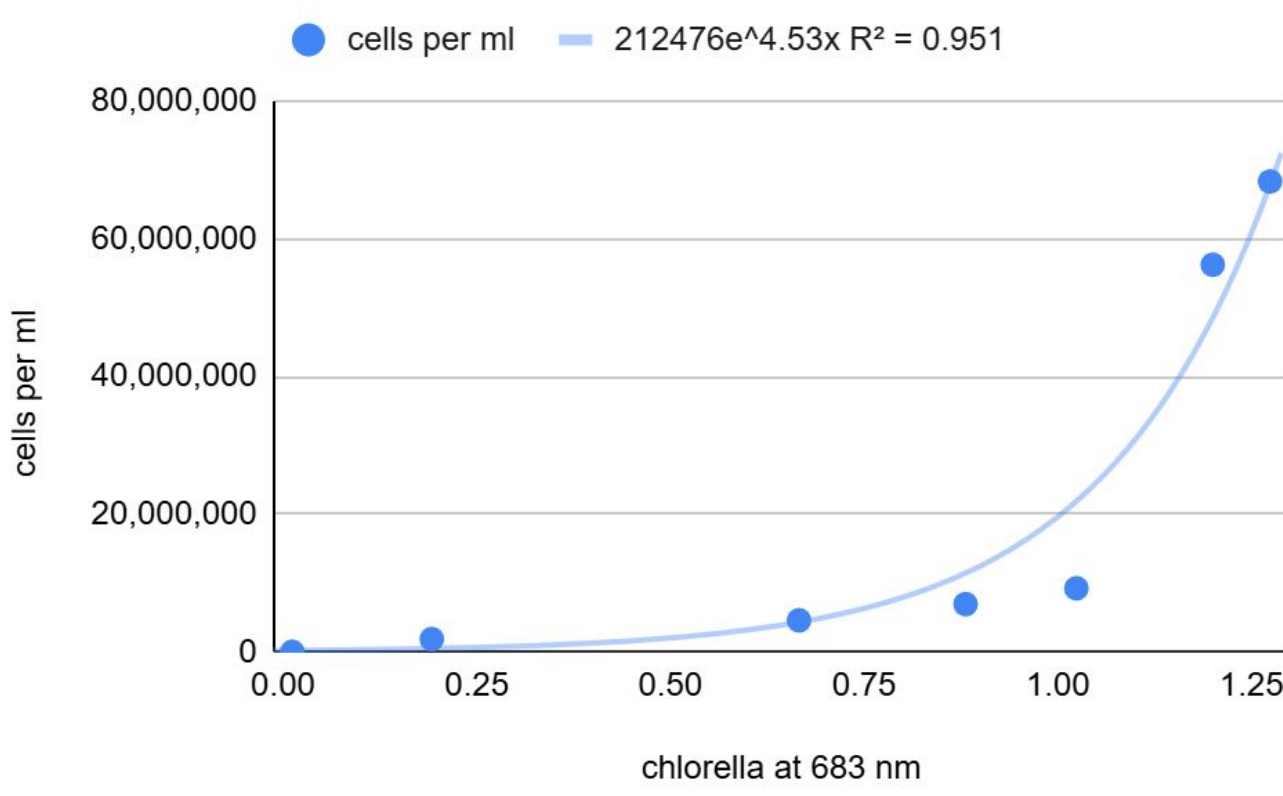


Figure 6: Cells per mL of chlorella vs the absorbance at 683 nm

## DEVELOPMENT OF DESIGN CONCEPT

Designing a biofertilizer from the algae produced from animal wastewater is an effective method to remove excess nutrients from water through algae consuming the nitrates and phosphates. Using naturally occurring processes such as flocculation (particles in a solution aggregate and form flocs) after the algae is harvested is an environmentally and cost friendly way to repurpose the separated algae, which can be dried and applied as fertilizer.

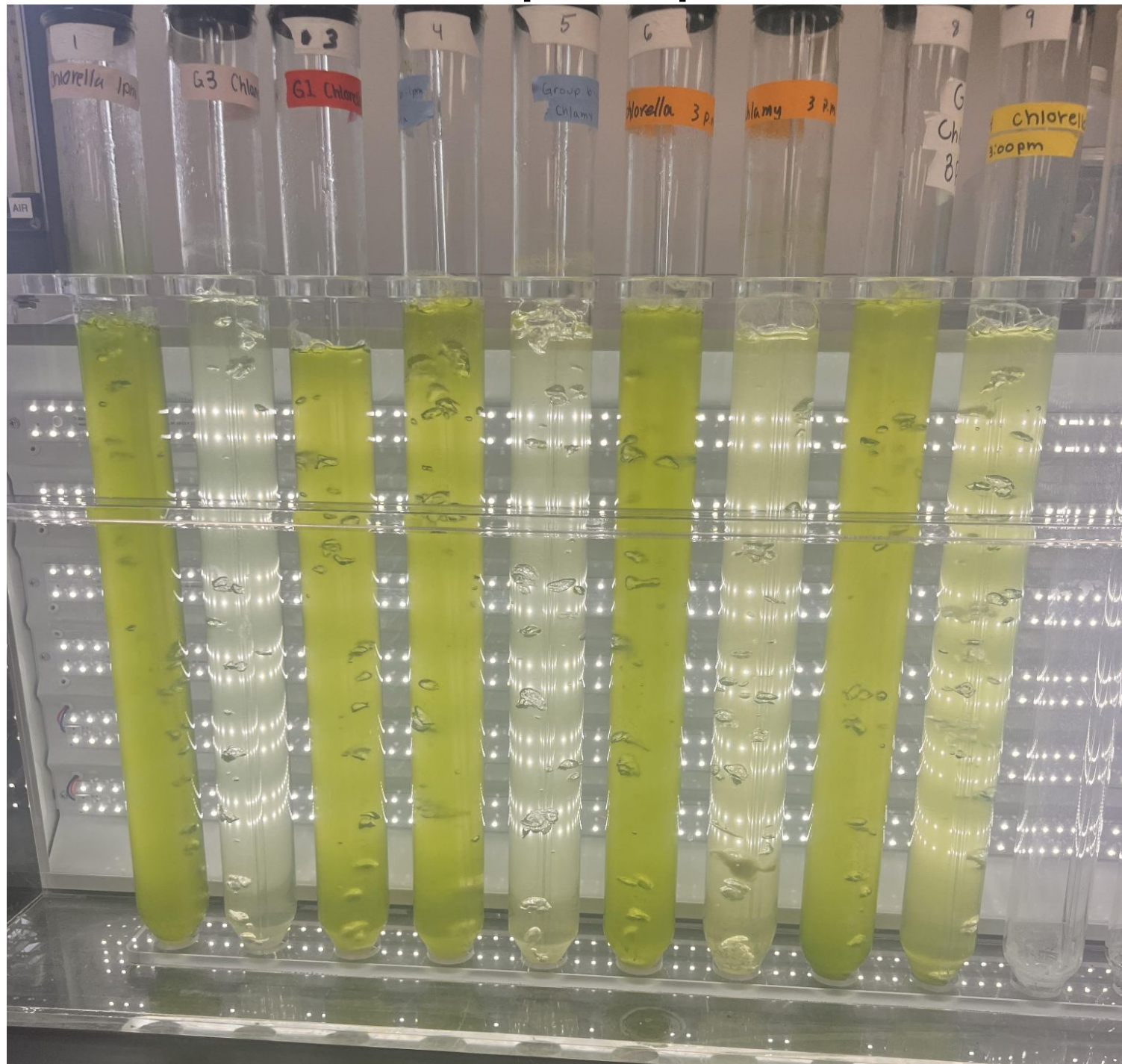


Figure 7: Vials of algae growing in the lab

## FUTURE CONSIDERATIONS

- Taking data on multiple types of algae to see if one is more efficient at removing nutrients. With one type, it is impossible to see if we chose the best type for our task.
- Taking more data in the lab- with more data, graphs will be more accurate and complete.
- Making our problem statement more narrow for the audience better understand our project.
- Researching better ways for harvesting and drying- more efficient and cost effective methods could be used.

## REFERENCES

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- Ma, F., Li, Y., Han, X., Li, K., Zhao, M., Guo, L., Li, S., Wang, K., Qin, K., Duan, J., Liu, Y., & Xu, Y. (n.d.). *Microalgae-based biofertilizer improves fruit yield and controls greenhouse gas emissions in a Hawthorn Orchard*. PLOS ONE.
- Ammar, E. E., Aioub, A. A. A., Elesawy, A. E., Karkour, A. M., Mouhamed, M. S., Amer, A. A., & EL-Sherashby, N. A. (2022). Algae as Bio-fertilizers: Between current situation and future prospective. *Saudi Journal of Biological Sciences*, 29(5), 3083–3096.
- Current status and perspectives of algae-based bioplastics: A reviewed potential for sustainability - sciencedirect. (n.d.-a).
- Figure 2: Standard specifications of biochemical properties of biofertilizer. | download scientific diagram. (n.d.-b).